1Kosmos ARR Dashboard

Complete Logic & Data Source Reference

Every metric, chart and calculation box explained

Table | Column | Formula | Business Meaning

Data Flow Overview

Excel upload (Booking Database.xlsx) -> Django parser -> BookingRecord table. Each row = one contract line. Key fields: current_arr (float), monthly_arr (JSON "YYYY-MM":value), end_user, business_unit, sales_person, sub_product_type, line_of_business. API endpoint: GET /api/dashboard/ -> React ARRDashboard.js renders all widgets below.

Sections in this document:

1. KPI Strip - 6 Headline Metric Cards

The six cards at the top are derived from _compute_ltm() in views.py and the sum of BookingRecord.current_arr. They compare current contract state against ARR values 12 months earlier.

KPI Card	Table	Column(s)	Formula	Meaning
Total ARR	BookingRecord	current_arr	SUM(current_arr) all filtered records	Closing ARR as of the latest month
LTM Change	BookingRecord	current_arr, monthly_arr	new_arr + upsell - churn - downsell	Net ARR movement over last 12 months
LTM New ARR	BookingRecord	current_arr, monthly_arr	SUM(current) where opening==0	ARR from brand-new logos in LTM
LTM Upsell	BookingRecord	current_arr, monthly_arr	SUM(curr-open) where curr>open>0	Expansion ARR from existing accounts
LTM Churn	BookingRecord	current_arr, monthly_arr	SUM(open) where open>0, curr==0	ARR lost to full cancellations
LTM Downsell	BookingRecord	current_arr, monthly_arr	SUM(open-curr) where 0<curr<open	ARR lost to partial contraction

How "opening_arr" is determined:

Backend finds latest month across all monthly_arr keys. Subtracts 12 months => ltm_start. For each BookingRecord, opening_arr = the monthly_arr value at or before ltm_start.

```
ltm_start = f"{year-1:04d}-{month:02d}"
past = sorted(k for k in monthly_arr.keys() if k <= ltm_start)
opening_arr = monthly_arr[past[-1]] if past else 0.0
current_arr = BookingRecord.current_arr
```

File: kosmos/kosmos_app/views.py lines 32-74 (_compute_ltm)

2. Metric Cards - ARR Growth, GRR%, NRR%

Metric	Table	Column(s)	Formula	Meaning
ARR Growth %	BookingRecord	current_arr, monthly_ar	ltm_change / opening_arr * 100	Percentage growth over last 12 months.
GRR%	BookingRecord	current_arr, monthly_ar	(opening - churn - downsell) / opening * 100	Retention EXCLUDING upsell. 100% = n
NRR%	BookingRecord	current_arr, monthly_ar	(opening - churn - downsell + upsell) / opening	Retention INCLUDING upsell. >100% me

views.py _compute_ltm() | React: ARRDashboard.js lines 188-190

3. Gauge Charts - GRR% and NRR% Visual Gauges

Two semi-circular gauges. Needle angle is proportional to the value against max scale (GRR max=120, NRR max=150). Values read from kpis.grr and kpis.nrr in the API response. No separate query - same _compute_ltm() values as section 2.

Table	BookingRecord
GRR formula	(opening - churn - downsell) / opening x 100
NRR formula	(opening - churn - downsell + upsell) / opening x 100
Scale (GRR)	0 - 120%
Scale (NRR)	0 - 150%

React: GaugeChart in arrShared.js | Data: kpis.grr, kpis.nrr

4. ARR by Customer - Treemap

Area chart: each rectangle = one customer, size = total current_arr. Top 15 customers.

Table	BookingRecord
Grouped by	end_user (falls back to bill_to if blank)

Value column	current_arr
Aggregation	SUM(current_arr) per customer, top 15 by value
API field	response.breakdowns.customers

```
label = end_user OR bill_to OR "Unspecified"
totals[label] += current_arr
Return top-15 sorted descending
```

React: Treemap in arrShared.js | Data: data.breakdowns.customers

5. ARR Flow - Sankey Diagram

Shows ARR flow: Opening -> New / Upsell / Renewals -> Churn -> Closing ARR. All values from the LTM computation (same as KPI strip).

Opening ARR	kpis.ltm_opening_arr
New ARR node	kpis.ltm_new_arr
Upsell node	kpis.ltm_upsell
Churn node	kpis.ltm_churn
Renewals node	opening_arr - churn - downsell (retained base)
Closing ARR	kpis.total_arr (SUM current_arr)

React: SankeyChart in arrShared.js | Data: kpis object

6. ARR Trend - Line Chart

Total closing ARR for every calendar month in the data. One point per month = SUM all contract ARR for that month.

Table	BookingRecord
Source column	monthly_arr (JSON: {"YYYY-MM": float})
Aggregation	For each month key in any record, SUM values for that month
API field	response.trend -> [{month, value}, ...]

```
trend_totals = defaultdict(float)
for record in records:
    for month, amount in record.monthly_arr.items():
        trend_totals[month] += float(amount or 0)
```

views.py lines 183-186 | React: LineChart in arrShared.js | Data: data.trend

7. Changes in ARR - Waterfall Chart by Month

Bar groups per month showing: New (green), Upsell (blue), Churn (red), Downsell (orange). Covers last 12 months of data.

Table	BookingRecord
Columns	monthly_arr (JSON)
Window	Last 13 months (gives 12 consecutive month-pairs)
API field	response.waterfall

```
For each month pair (prev_m -> curr_m):
  curr = monthly_arr.get(curr_m, 0)
  prev = monthly_arr.get(prev_m, 0)

  if prev == 0 and curr > 0:   new      += curr
  elif prev > 0 and curr == 0: churn    += prev
  elif curr > prev:            upsell   += curr - prev
  elif curr < prev:            downsell += prev - curr
```

views.py _build_waterfall() lines 77-97 | React: WaterfallChart in arrShared.js

8. ARR by Business Unit - Bar Chart

Grouped by business_unit, bar height = total current ARR. Computed client-side.

Table	BookingRecord (via response.records)
Group field	business_unit
Value	monthly_arr[latestMonth] OR current_arr
Aggregation	SUM per business_unit, sorted descending
Also shows	LTM New+Upsell, LTM Churn+Downsell, NRR%, contract count per BU

React ARRDashboard.js line 130 | buildGroupStats(records, "business_unit", ...)

9. ARR by Sales Rep - Bar Chart

Same logic as Business Unit chart but grouped by sales_person.

Table	BookingRecord (via response.records)
Group field	sales_person
Value	monthly_arr[latestMonth] OR current_arr
Aggregation	SUM per sales_person, sorted descending

React ARRDashboard.js line 131 | buildGroupStats(records, "sales_person", ...)

10. New ARR by Quarter - Bar Chart

Groups waterfall new+upsell into fiscal quarters. One bar per quarter.

Source data	response.waterfall (month-level from server)
Table behind	BookingRecord.monthly_arr
Aggregation	quarterLabel(month) groups months into Q1/Q2/Q3/Q4 YYYY
Value	SUM(waterfall.new + waterfall.upsell) per quarter
Max bars	Last 8 quarters shown

```
qtrMap = {}
waterfall.forEach(row => {
    q = quarterLabel(row.month)  // "Q1 2024"
    qtrMap[q] = (qtrMap[q]||0) + row.new + row.upsell
})
```

React ARRDashboard.js lines 135-139

11. Churn & Downsell Trend - Mini Movement Bars

Compact stacked bar per month: New (green), Upsell (blue), Churn (red), Downsell (orange). Last 12 months. Bar width proportional to value vs max component.

Source data response.waterfall (last 12 entries)

Table behind BookingRecord.monthly_arr

Scale Each segment width = (component / max_any_component) * 100%

React: MiniMovementBars in ARRDashboard.js lines 81-98 | Data: waterfall.slice(-12)

12. Sub-Product ARR - Table

Lists every sub_product_type with ARR metrics. Computed client-side.

Table BookingRecord (via response.records)

Group field sub_product_type

Column Shown	Calculation
Sub-Product	sub_product_type field from BookingRecord
Current ARR	SUM(monthly_arr[latestMonth]) per product group
LTM New	SUM(newArr + upsell) - new logos + expansion in LTM
Churn/Downsell	SUM(churn + downsell) - ARR lost in LTM window
Share %	product_current_arr / kpis.total_arr * 100

React ARRDashboard.js lines 270-299 | buildGroupStats("sub_product_type")

13. ARR by Line of Business - LOB Cards

Grid of up to 6 cards, one per LOB. Hover reveals detail overlay.

Table BookingRecord (via response.records)

Group field line_of_business

Metric on Card	Formula
ARR Value	SUM(monthly_arr[latestMonth]) for all contracts in this LOB
% of Total	lob.value / kpis.total_arr * 100
LTM New	SUM(newArr + upsell) within LTM for this LOB
LTM Churn	SUM(churn + downsell) within LTM for this LOB
NRR%	(opening - churn - downsell + upsell) / opening * 100 per LOB
Contracts	Count of BookingRecord rows in this LOB

React ARRDashboard.js lines 301-347

14. Customer ARR Detail - Sortable Table

Full customer-level table from _build_customer_360(). Up to 100 rows shown. Click row to expand opening ARR and LTM change detail.

Table BookingRecord (via response.customer_360)

Grouped by end_user OR bill_to

Column	Source & Calculation
Customer	end_user OR bill_to OR "Unspecified"
Current ARR	SUM(current_arr) across all contracts for this customer

LTM Change	current_arr_total - opening_arr_total
LTM %	ltm_change / opening_arr * 100
BU	business_unit of first contract found
Sales Rep	sales_person of first contract
Contracts	Count of BookingRecord rows for this customer
Products	Distinct set of sub_product_type values
Trend Sparkline	Last 12 months of monthly_arr summed per month
Opening ARR (expand)	monthly_arr value at or before ltm_start

views.py _build_customer_360() lines 100-137 | React ARRDashboard.js lines 349-424

15. LTM Calculation Engine - Core Algorithm

Central algorithm powering almost every number in the dashboard. Runs in views.py _compute_ltm() and is replicated client-side in buildGroupStats().

Step 1 - Determine time window

```
# Find the latest month across all records
latest_month = max(all monthly_arr keys)

# LTM start = exactly 12 months before latest
ltm_start = f"{year-1:04d}-{month:02d}"
```

Step 2 - Classify each BookingRecord

```
for record in records:
    monthly = record.monthly_arr       # JSON {"YYYY-MM": float}
    curr    = record.current_arr       # Current ARR

    # Opening = last known value at or before ltm_start
    past = sorted(k for k in monthly.keys() if k <= ltm_start)
    prev = monthly[past[-1]] if past else 0.0

    opening_total += prev

    if prev == 0 and curr > 0:    new_arr  += curr
    elif prev > 0 and curr == 0:  churn    += prev
    elif curr > prev:             upsell   += curr - prev
    elif curr < prev:             downsell += prev - curr
```

Step 3 - Compute retention metrics

```
ltm_change = new_arr + upsell - churn - downsell
GRR = max(0, (opening - churn - downsell) / opening * 100)
NRR = max(0, (opening - churn - downsell + upsell) / opening * 100)
growth_pct = ltm_change / opening * 100
```

views.py lines 32-74 (server) | ARRDashboard.js buildGroupStats() lines 39-79 (client)

Classification Decision Tree

Condition	Category	Amount Added
opening == 0 AND current > 0	New ARR	current_arr
opening > 0 AND current == 0	Churn	opening_arr
current > opening > 0	Upsell (Expansion)	current - opening
0 < current < opening	Downsell (Contraction)	opening - current
current == opening	Renewal / Flat	0 (no movement counted)

16. Database Model Reference

The ARR Dashboard reads exclusively from BookingRecord. Full field reference below.

Table: BookingRecord (kosmos_app_bookingrecord)

Field	Type	Purpose & Dashboard Usage
key_id	CharField	Internal row identifier from source Excel
entity	CharField	1Kosmos legal entity
currency	CharField	Deal currency code
contract_id	CharField	Contract identifier
contract_name	CharField	Contract display name
sales_person	CharField	Owning sales rep <- used in ARR by Sales Rep
business_unit	CharField	Business unit <- used in ARR by BU
industry	CharField	Customer industry
bill_to	CharField	Billing customer name
end_user	CharField	End-user name <- primary customer key
product_type	CharField	High-level product category
sub_product_type	CharField	Sub-product <- used in Sub-Product table
revenue_type	CharField	"recurring" vs "non-recurring"
tcv_usd	FloatField	Total Contract Value in USD
arr_usd	FloatField	ARR in USD from source sheet
booking	FloatField	Booking amount
term_start	DateField	Contract start date
term_end	DateField	Contract end date
line_of_business	CharField	LOB <- used in ARR by LOB cards
current_arr	FloatField	*** PRIMARY: current closing ARR for this contract
monthly_arr	JSONField	*** KEY: {"YYYY-MM": float} - ARR per month history
attribute1-15	CharField	Custom/flexible attributes from source file

Supporting Tables (ARR Dashboard does NOT use these)

ARAgingRecord, ARPaymentRecord, ARRenewalRecord -> AR Dashboard tab only. PipelineRecord -> Pipeline tab only.

17. Quick Reference - All Widgets at a Glance

Widget / Chart	DB Table	Key Column(s)	Computed In
Total ARR KPI	BookingRecord	current_arr	views.py (server)
LTM Change KPI	BookingRecord	current_arr, monthly_arr	views.py _compute_ltm
LTM New ARR KPI	BookingRecord	current_arr, monthly_arr	views.py _compute_ltm
LTM Upsell KPI	BookingRecord	current_arr, monthly_arr	views.py _compute_ltm
LTM Churn KPI	BookingRecord	current_arr, monthly_arr	views.py _compute_ltm
LTM Downsell KPI	BookingRecord	current_arr, monthly_arr	views.py _compute_ltm
ARR Growth % Card	BookingRecord	current_arr, monthly_arr	views.py _compute_ltm
GRR% Metric Card	BookingRecord	current_arr, monthly_arr	views.py _compute_ltm
NRR% Metric Card	BookingRecord	current_arr, monthly_arr	views.py _compute_ltm
GRR% Gauge	BookingRecord	current_arr, monthly_arr	views.py (same value)
NRR% Gauge	BookingRecord	current_arr, monthly_arr	views.py (same value)
ARR by Customer Treemap	BookingRecord	current_arr, end_user	views.py breakdown()
ARR Flow Sankey	BookingRecord	current_arr, monthly_arr	views.py _compute_ltm
ARR Trend Line Chart	BookingRecord	monthly_arr	views.py trend_totals loop
Waterfall Chart	BookingRecord	monthly_arr	views.py _build_waterfall
ARR by Business Unit	BookingRecord	business_unit, monthly_arr	React buildGroupStats
ARR by Sales Rep	BookingRecord	sales_person, monthly_arr	React buildGroupStats
New ARR by Quarter	BookingRecord	monthly_arr (via waterfall)	React quarterLabel()
Churn Trend Mini Bars	BookingRecord	monthly_arr (via waterfall)	React MiniMovementBars
Sub-Product ARR Table	BookingRecord	sub_product_type, monthly_arr	React buildGroupStats
LOB Cards	BookingRecord	line_of_business, monthly_arr	React buildGroupStats
Customer Detail Table	BookingRecord	end_user, current_arr, monthly_arr	views.py _build_customer_360

Key principle: monthly_arr is the single source of truth for all time-series and LTM calculations. current_arr is the snapshot value used for total sums. Every movement metric derives from comparing these two fields.